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T11 - 05

Assignment No. 2

AIM – Write a program in typescript to implement Simple Calculator

LO2 - Understand the basic concepts of Typescript for designing web applications

THEORY:

TypeScript is a typed superset of JavaScript developed by Microsoft. It adds static typing and object-oriented features to JavaScript, helping developers catch errors at compile time and write cleaner, scalable code.

Key Features of TypeScript:

1. Static Typing – Types are checked during development.
2. Interfaces & Classes – Supports object-oriented programming.
3. Type Inference – Automatically infers variable types.
4. Cross-platform – Compiles to plain JavaScript.

Implementing a Simple Calculator in TypeScript

◇ Features:

- Perform basic operations: Add, Subtract, Multiply, Divide.
- Take two numbers and an operator.

PROGRAM:

```
App.jsx import './App.css' import
Calculator from
'./components/calculator' function App() { return (
  <>
    <Calculator/>
  </>
)
} export default
App
```

```
Calculator.jsx import React, { useState } from
"react"; import "./Calculator.css"; const Calculator:
React.FC = () => { const [input, setInput] =
useState<string>("");

  const handleClick = (value: string) => { setInput((prev) =>
prev + value);
  };

  const handleClear = () => { setInput("");
  };

  const handleCalculate = () => {
```

```

    try {      setInput(eval(input).toString()); // Using eval (safe for simple
calculations)    } catch {      setInput("Error");
  }
};

```

```

return (
  <div className="calculator">
    <input type="text" className="calculator-screen" value={input}
readOnly />
    <div className="buttons">
      {[ "7", "8", "9", "/" ].map((btn) => (
        <button key={btn} onClick={() => handleClick(btn)}>{btn}</button>
      ))}
      {[ "4", "5", "6", "*" ].map((btn) => (
        <button key={btn} onClick={() => handleClick(btn)}>{btn}</button>
      ))}
      {[ "1", "2", "3", "-" ].map((btn) => (
        <button key={btn} onClick={() => handleClick(btn)}>{btn}</button>
      ))}
      {[ "0", ".", "+", "=" ].map((btn) =>
btn === "=" ? (
        <button key={btn} onClick={handleCalculate} className="equal">
          {btn}
        </button>
      ) : (

```

```
    <button key={btn} onClick={() => handleClick(btn)}>
      {btn}
    </button>
  )
  )}
  <button onClick={handleClear} className="clear">C</button>
</div>
</div>
);
}; export default
Calculator;
```

```

JS calculator.js > calculator
1  var _a;
2  var add = function (a, b) { return a + b; };
3  var sub = function (a, b) { return a - b; };
4  var mul = function (a, b) { return a * b; };
5  var div = function (a, b) {
6      if (b === 0) {
7          throw new Error("Division by zero is not allowed.");
8      }
9      return a / b;
10 };
11 function calculator(a, b, operation) {
12     switch (operation) {
13         case "add":
14             return add(a, b);
15         case "sub":
16             return sub(a, b);
17         case "mul":
18             return mul(a, b);
19         case "div":
20             return div(a, b);
21         default:
22             throw new Error("Invalid operation. Supported operations are: add, sub, mul, div.");
23     }
24 }
25 (_a = document.getElementById("calculate")) === null || _a === void 0 ? void 0 : _a.addEventListener("click", function () {
26     var num1 = parseFloat(document.getElementById("num1").value);
27     var num2 = parseFloat(document.getElementById("num2").value);
28     var operation = document.getElementById("operation").value;
29     var resultElement = document.getElementById("result");

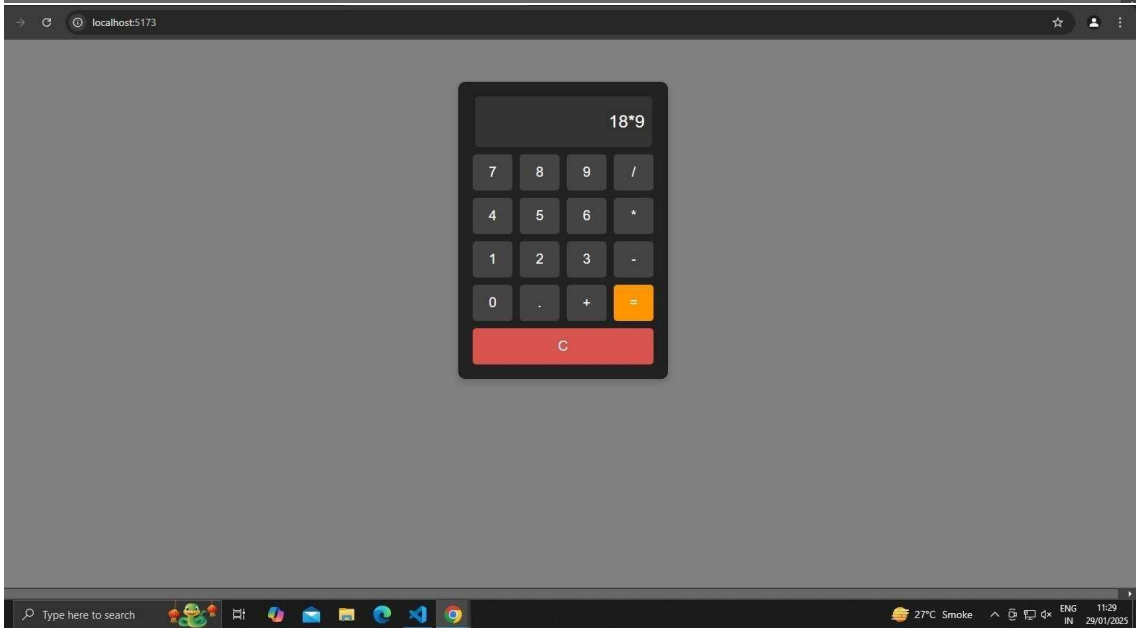
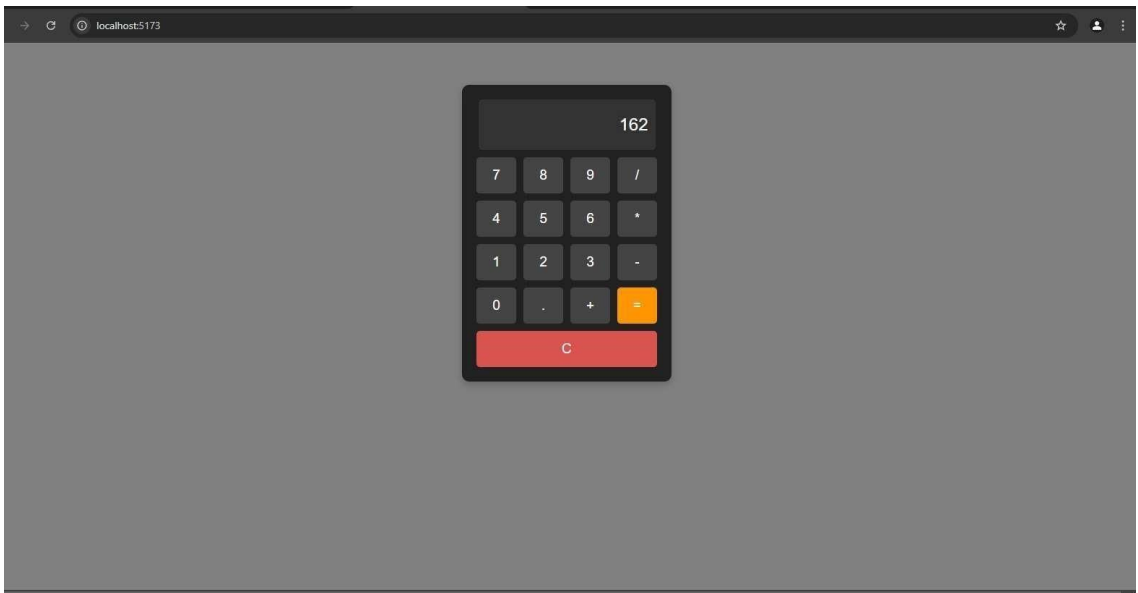
```

```

JS calculator.js > calculator
22         throw new Error("Invalid operation. Supported operations are: add, sub, mul, div.");
23     }
24 }
25 (_a = document.getElementById("calculate")) === null || _a === void 0 ? void 0 : _a.addEventListener("click", function () {
26     var num1 = parseFloat(document.getElementById("num1").value);
27     var num2 = parseFloat(document.getElementById("num2").value);
28     var operation = document.getElementById("operation").value;
29     var resultElement = document.getElementById("result");
30     try {
31         var result = calculator(num1, num2, operation);
32         if (resultElement) {
33             resultElement.textContent = "Result: ".concat(result);
34         }
35     }
36     catch (error) {
37         if (resultElement) {
38             resultElement.textContent = "Error: ".concat(error.message);
39         }
40     }
41 });
42

```

OUTPUT:





Calculator

Add (+)

Calculate

Clear

Result will appear here

Calculator

Add (+)

Add (+)

Subtract (-)

Multiply (x)

Divide (/)

Calculator

Subtract (-)

Calculate

Clear

Result: 13

